Movie Recommendation System

Online streaming platforms have plenty of movies in their repository and a Recommendation System would improve customer satisfaction and the revenue of the platform.

Build various recommendation systems such as Knowledge/Rank and User/Item based Collaborative Filtering Recommendation Systems.

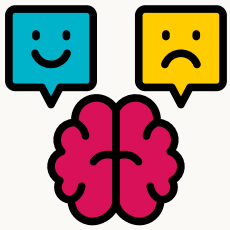
Python | Jupyter Notebook

1. Overview and exploration of the dataset
2. Create Rank-Based Recommendation System
3. User based Collaborative Filtering Recommendation System
4. Item based Collaborative Filtering Recommendation System
5. Based Collaborative Filtering - Matrix Factorization using SVD
6. Comparing the rating predictions for already interacted movies

DATA INSIGHTS

1. User based RMSE is 0.9925 and the item based RMSE is 1.0032.
2. Collaborative Models are in charge of using the user-item-ratings data in order to find similarities and make predictions.

3. So, they are more focused on understanding the patterns and obtaining conclusions from behavior to do better predictions and recommendations.



LESSONS & PERSPECTIVES

1. Tuning SVD matrix factorization model somewhat improved the base line SVD (tuned user based of 0.9908 and tuned item based of 0.9395).
2. Matrix Factorization describes their properties and recommend an item considering the proximity to the user, which in some cases is helpful for using patterns to obtain conclusions of users with similar characteristics.

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